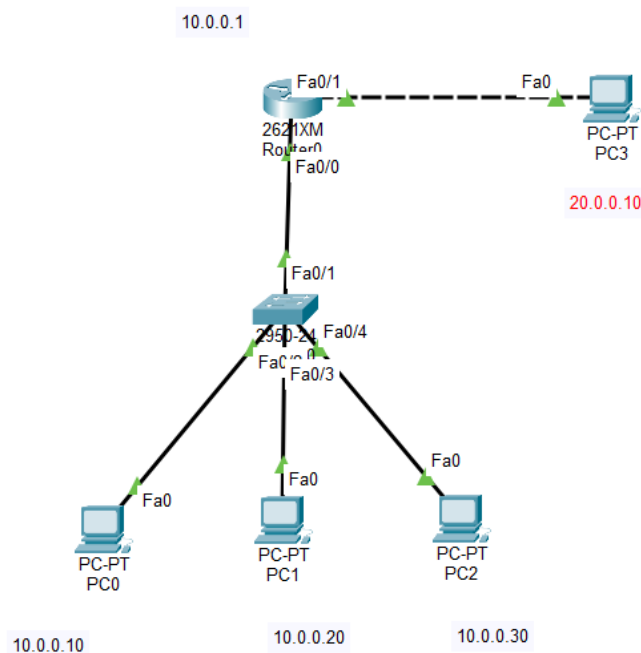


3.NAT and PAT ConceptsTheory :What is NAT/PAT? Importance in IP address conservation.Practical :Configure NAT and PAT on a router.Task: Simulate external connectivity using NAT/PAT.

Task 1.

In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.



```
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 20.0.0.10

Pinging 20.0.0.10 with 32 bytes of data:

Reply from 20.0.0.10: bytes=32 time<1ms TTL=127
Reply from 20.0.0.10: bytes=32 time<1ms TTL=127
Reply from 20.0.0.10: bytes=32 time<1ms TTL=127
Reply from 20.0.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 20.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Task 2.

Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.

```
Building configuration...
[OK]
Router(config)#access-list 1 permit 10.0.0.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface Fa0/1 overload
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Task 3.

Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

```
Router>
Router>show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 20.0.0.1:17        10.0.0.10:17      20.0.0.10:17      20.0.0.10:17
icmp 20.0.0.1:18        10.0.0.10:18      20.0.0.10:18      20.0.0.10:18
icmp 20.0.0.1:19        10.0.0.10:19      20.0.0.10:19      20.0.0.10:19
icmp 20.0.0.1:20        10.0.0.10:20      20.0.0.10:20      20.0.0.10:20
_
```

Task 4.

Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

```
Router>show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 20.0.0.1:1024      10.0.0.30:1       20.0.0.10:1       20.0.0.10:1024
icmp 20.0.0.1:1025      10.0.0.30:2       20.0.0.10:2       20.0.0.10:1025
icmp 20.0.0.1:1026      10.0.0.30:3       20.0.0.10:3       20.0.0.10:1026
icmp 20.0.0.1:1027      10.0.0.30:4       20.0.0.10:4       20.0.0.10:1027
icmp 20.0.0.1:1         10.0.0.20:1       20.0.0.10:1       20.0.0.10:1
icmp 20.0.0.1:21        10.0.0.10:21      20.0.0.10:21      20.0.0.10:21
icmp 20.0.0.1:22        10.0.0.10:22      20.0.0.10:22      20.0.0.10:22
icmp 20.0.0.1:23        10.0.0.10:23      20.0.0.10:23      20.0.0.10:23
icmp 20.0.0.1:24        10.0.0.10:24      20.0.0.10:24      20.0.0.10:24
icmp 20.0.0.1:2         10.0.0.20:2       20.0.0.10:2       20.0.0.10:2
icmp 20.0.0.1:3         10.0.0.20:3       20.0.0.10:3       20.0.0.10:3
icmp 20.0.0.1:4         10.0.0.20:4       20.0.0.10:4       20.0.0.10:4
```

Task 5.

Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

NAT

NAT (Network Address Translation) converts private IP addresses into public IP addresses.

Private IP	Public IP
10.0.0.10	20.0.0.100
10.0.0.20	20.0.0.101

Without NAT, private IP addresses cannot normally be directly routed across the public Internet.

PAT

PAT (Port Address Translation) goes one step further. Many private devices can use one public IP address.

```
10.0.0.10 —┐
10.0.0.20 —┴—▶ 20.0.0.100
10.0.0.30 —┘
```

This is extremely useful for large applications such as Zomato or Flipkart, because millions of users' devices can use private addresses inside their networks while NAT/PAT conserves the limited supply of IPv4 public addresses.

scenario 1 — Wrong inside/outside configuration

If the router interface connected to the internal network is not configured as:

```
ip nat inside
```

or the external interface is not configured as:

```
ip nat outside
```

NAT may not work correctly, causing users to lose Internet connectivity.

scenario 2 — Incorrect ACL

Suppose the administrator accidentally writes:

```
access-list 1 permit 10.0.1.0 0.0.0.255
```

while the actual users are on:

```
10.0.0.0/24
```

The users won't match the NAT rule, so their traffic won't be translated and they may not be able to access external servers.